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CONNECTOR

Abstract

A shell block of a connector has a through hole and a receiving portion. The through hole pierces the shell block and has a large aperture portion and a small aperture portion. The large aperture portion is positioned forward of the small aperture portion and communicates with the small aperture portion. The small aperture portion has a first size as a shortest size. The receiving portion is located at a boundary between the large aperture portion and the small aperture portion and faces forward. An insulator member of a contact assembly has a held portion, and the contact assembly is divided by the held portion into a front portion and a rear portion. The held portion is brought into contact with the receiving portion in the through hole. The rear portion has a second size as a tallest size. The second size is smaller than the first size.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is based on and claims the benefit of U.S. Provisional Patent Application No. 63/562,362, filed on Mar. 7, 2024, which is incorporated herein by reference in its entirety, and also claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. JP2024-209144 filed Nov. 29, 2024, the contents of which are incorporated herein in their entirety by reference.

BACKGROUND OF THE INVENTION

[0002] This invention relates to a connector, particularly, to a connector which is provided with a contact assembly and a shell block.

[0003] JP2024-516208A (Patent Document 1) discloses an example of a connector which is provided with a contact assembly and a shell block.

[0004] Referring to FIG. 18, a connector 90 disclosed in Patent Document 1 is provided with a contact assembly 92 and a shell block 94. The contact assembly 92 is attached to the shell block 94 and accommodated in a case 96 of an electronic device together with the shell block 94.

[0005] As shown in FIG. 18, the contact assembly 92 is formed with a protruding portion 921 protruding in directions perpendicular to a front-rear direction. On the other hand, the shell block 94 is formed with a stopper 941. The stopper 941 at least partly defines a hole 943 passing through the shell block 94 in the front-rear direction. The contact assembly 92 is inserted into the hole 943 from a rear of the shell block 94 and attached to the shell block 94. The protruding portion 921 of the contact assembly 92 is brought into abutment with the stopper 941 of the shell block 94, so that the contact assembly 92 is positioned in the front-rear direction with respect to the shell block 94.

[0006] As shown in FIG. 18, in the connector 90, a rear end portion of the contact assembly 92 is covered with a resin layer 98 at least in part. The resin layer 98 prevents moisture from entering the interior of the electronic device through internal gaps of the contact assembly 92 or a gap between the contact assembly 92 and the shell block 94.

SUMMARY OF THE INVENTION

[0007] With increasing density of parts in an electronic device and increasing in size of built-in battery, there is a demand to reduce a length size of a connector which is an external interface for the electronic device.

[0008] It is an object of the present invention to provide a connector having a reduced length size.

[0009] One aspect of the present invention provides a connector which is provided with a shell block and a contact assembly. The shell block has a through hole and a receiving portion. The through hole pierces the shell block in a front-rear direction and has a large aperture portion and a small aperture portion. The large aperture portion is located forward of the small aperture portion in the front-rear direction and communicates with the small aperture portion. The large aperture portion is larger than the smaller aperture portion in a plane perpendicular to the front-rear direction. The through hole has a first size as a size of a shortest part of the small aperture portion in an up-down direction perpendicular to the front-rear direction. The receiving portion is provided on a boundary between the large aperture portion and the small aperture portion in the through hole and faces forward in the front-rear direction. The contact assembly comprises at least one contact and an insulator member which fixes and holds the contact. The insulator member has a held

portion projecting in directions perpendicular to the front-rear direction. The contact assembly is divided by the held portion into a front portion located forward of the held portion and a rear portion located rearward of the held portion. The held portion is brought into contact with the receiving portion in the through hole. The contact assembly has a second size as a size of a tallest part of the rear portion in the up-down direction. The second size is smaller than the first size. [0010] The connector according to the above-mentioned aspect is formed so that the contact assembly is attached to the shell block from a front of the shell block. This structure can reduce, in the front-rear direction (or a length direction), a size necessary to cover a rear end portion of the contact assembly with a potting resin, so that a size of the rear portion of the connector can be smaller than that of a conventional one. [0011] An appreciation of the objectives of the present invention and a more complete understanding of its structure may be had by studying the following description of the preferred embodiment and by referring to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a perspective view partly showing an electronic device provided with a connector according to an embodiment of the present invention. The connector is depicted through a case of the electronic device. A circuit board on which the connector is mounted is provided with a relay connector.

[0013] FIG. 2 is another perspective view showing the electronic device of FIG. 1. A relay flexible printed cable is connected to the relay connector on the circuit board.

[0014] FIG. 3 is a perspective view showing the connector according to the embodiment of the present invention.

[0015] FIG. 4 is an exploded, bottom, perspective view showing the connector of FIG. 3. A potting resin is omitted.

[0016] FIG. 5 is an exploded, top, perspective view showing the connector of FIG. 3.

[0017] The potting resin is omitted.

[0018] FIG. 6 is a cross-sectional, perspective view showing the connector of FIG. 3. A back shell is omitted.

[0019] FIG. 7 is a vertical, cross-sectional view showing the connector of FIG. 6.

[0020] FIG. 8 is a perspective view showing a contact assembly and a fixing plate which are included in the connector of FIG. 3.

[0021] FIG. 9 is a perspective view showing the contact assembly, the fixing plate, a shell block and a waterproof gasket which are included in the connector of FIG. 3. The fixing plate is attached to the contact assembly. The waterproof gasket is attached to the shell block.

[0022] FIG. 10 is a vertical, cross-sectional view showing the contact assembly, the fixing plate and the shell block which are included in the connector of FIG. 3.

[0023] FIG. 11 is a perspective view showing a modified example of the connector of FIG. 3. The connector is mounted on a circuit board.

[0024] FIG. 12 is a perspective view partially showing an electronic device provided with the connector of FIG. 11. The connector is depicted through a case of the electronic device.

[0025] FIG. 13 is a vertical, cross-sectional view showing another modified example of the connector of FIG. 3.

[0026] FIG. 14 is a vertical, cross-sectional view showing the connector of FIG. 13. The connector is mounted on a circuit board.

[0027] FIG. 15 is a front view showing the shell block and the waterproof gasket which are included in the connector of FIG. 5.

[0028] FIG. **16** is a front view showing a modified example of the shell block of FIG. **15** and the waterproof gasket.

[0029] FIG. **17** is a front view showing another modified example of the shell block of FIG. **15** and the waterproof gasket.

[0030] FIG. **18** is a cross-sectional view showing a connector disclosed in Patent Document 1.

[0031] While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION

[0032] As understood from FIGS. **1** and **2**, a connector **1** shown is provided in a case **73** of an electronic device when used and mounted on a circuit board **71** provided in the case **73**. A board-to-FPC connector **75** is mounted on the circuit board **71**, and a relay flexible printed cable (FPC) **77** is connected to the board-to-FPC connector **75**. The connector **1** is detachable with respect to a mating connector (not shown) along a front-rear direction. In the present embodiment, the front-rear direction coincides with a length direction. In the figures, the front-rear direction is an X-direction. A positive X-direction is directed forward while a negative X-direction is directed rearward. The connector **1** shown is a waterproof connector. However, the present invention is also applicable to connectors other than the waterproof connector.

[0033] As shown in FIG. **3**, the connector **1** is provided with a shell block **10** and a contact assembly **20**. As shown in FIGS. **3** to **6**, the connector **1** is further provided with a fixing plate **30**, a waterproof gasket **40**, a potting resin **50** and a back shell **60**.

[0034] As understood from FIG. **5**, the shell block **10** is made by a manufacturing process, such as metal molding or cutting, etc., and it is a structure having thickness elements of a plurality of kindness. The shell block **10** has a through hole **110** and a receiving portion **150**. As shown in FIGS. **11** and **12**, a shell block **10A** of a connector **1A** of a modified example may be further provided with a fixed portion **170** which is fixed to the case **73** of the electronic device by the use of screws **79**.

[0035] As shown in FIG. **5**, the through hole **110** pierces the shell block **10** in the front-rear direction and has a large aperture portion **120** and a small aperture portion **130**. The large aperture portion **120** is positioned forward of the small aperture portion **130** in the front-rear direction and communicates with the small aperture portion **130**. The large aperture portion **120** is larger than the smaller aperture portion **130** in a plane perpendicular to the front-rear direction. In the figure, the plane perpendicular to the front-rear direction is a Y-Z plane. In detail, as shown in FIG. **15**, in the through hole **110** shown, the large aperture portion **120** is larger than the small aperture portion **130** in each of an up-down direction perpendicular to the front-rear direction and a pitch direction perpendicular to both the front-rear direction and the up-down direction. In the figure, the up-down direction is a Z-direction, and the pitch direction is a Y-direction. A positive Z-direction is directed upward while a negative Z-direction is directed downward. However, the present invention is not limited thereto. As a modified example of FIG. **16**, a large aperture portion **120A** may be larger than a small aperture portion **130A** only in the pitch direction. As a modified example of FIG. **17**, a large aperture portion **120B** may be larger than a small aperture portion **130B** only in the up-down direction.

[0036] As understood from FIGS. **5** and **15**, a step is formed at a boundary between the large aperture portion **120** and the small aperture portion **130**. The receiving portion **150** is formed on the step. In detail, the receiving portion **150** is provided on the boundary between the large aperture portion **120** and the small aperture portion **130** in the through hole **110** and faces forward in the front-rear direction. In the shell block **10** shown, the receiving portion **150** has a ring shape.

However, the present invention is not limited thereto. For example, in the shell block **10A** of FIG. **16**, receiving portions **150A** are positioned only at both ends in the pitch direction. Moreover, in a shell block **10B** of FIG. **17**, receiving portions **150B** are positioned only at both ends in the up-down direction.

[0037] As shown in FIG. **7**, the through hole **110** has a first size **S1** as a size of a shortest part of the small aperture portion **130** in the up-down direction. In other words, a smallest size of the through hole **110** is the first size **S1** in the up-down direction.

[0038] As understood from FIGS. **3** to **5**, the waterproof gasket **40** is attached to the shell block **10** so that the waterproof gasket **40** surrounds the shell block **10**. In detail, the waterproof gasket **40** surrounds a front end of the shell block **10** and vicinity of the front end. The waterproof gasket **40** shown is made of elastic material, such as rubber. The waterproof gasket **40** has a ring shape.

[0039] As shown in FIG. **5**, the contact assembly **20** is provided with contacts **210** and an insulation member **220** which fixes and holds the contacts **210**.

[0040] As understood from FIG. **5**, the insulation member **220** has a held portion **222** projecting in directions perpendicular to the front-rear direction. The held portion **222** shown is a flange and projects in a plane perpendicular to the front-rear direction. The contact assembly **20** is divided by the held portion **222** into a front portion **23** located forward of the held portion **222** and a rear portion **25** located rearward of the held portion **222**. In particular, the front portion **23** shown has a tongue-like shape.

[0041] As shown in FIG. **7**, the contact assembly **20** has a second size **S2** as a size of a tallest part of the rear portion **25** in the up-down direction. In other words, a largest size of the rear portion **25** of the contact assembly **20** is the second size **S2** in the up-down direction. The second size **S2** is smaller than the first size **S1**.

[0042] As shown in FIG. **5**, the contacts **210** are partly embedded in the insulation member **220** by insert molding and held by the insulation member **220**. The contacts **210** shown are plural in number and arranged in two, upper and lower, rows. Furthermore, the contacts **210** in each of the rows are arranged in the pitch direction.

[0043] As understood from FIGS. **5** to **7**, the contact **210** has a contact portion **212** and a connection portion **214**. The contact portion **212** is located at the front portion **23** of the contact assembly **20**. The contact portion **212** shown is exposed upward or downward of the front portion **23**. The contact portion **212** is brought into contact with a mating contact (not shown) with which the mating connector (not shown) is provided in a connection state of the connector **1** and the mating connector. The connection portion **214** is located at the rear portion **25** of the contact assembly **20**. In the connector **1** shown, the rear portion **25** is formed of only parts of the contacts **210**. However, the present invention is not limited thereto. The rear portion **25** may include a part of the insulation member **220**.

[0044] As understood from FIGS. **1**, **3** and **5**, when the connector **1** is mounted on the circuit board **71**, the circuit board **71** is arranged under the rear portion **25** of the contact assembly **20**. The connection portion **214** of the contact **210** is connected to a conductor (not shown) on the circuit board **71** when the connector **1** is mounted on the circuit board **71**. In detail, the connection portion **214** shown is soldered to the conductor on the circuit board **71**. However, the present invention is not limited thereto. For example, as understood from FIGS. **13** and **14**, a connection portion **214A** of a modified example may be supported by a supporting portion **216A**, which is resiliently deformable, and brought into contact with the conductor on the circuit board **71** by using resilient deformation of the supporting portion **216A**.

[0045] As shown in FIG. **7**, the shell block **10** holds the insulation member **220** of the contact assembly **20** by using the fixing plate **30**. Specifically, the fixing plate **30** is fixed to the shell block **10**. Furthermore, the held portion **222** is sandwiched between the fixing plate **10** and the receiving portion **150** in the front-rear direction. Accordingly, the held portion **222** is fixed in position in the front-rear direction. In the state, the held portion **222** is brought into contact with the receiving

portion **150** in the through hole **110**. As an alternative for holding the insulation member **220** by the shell block **10**, it is conceived to press-fit the held portion **222** into the through hole **110** or thermally weld the held portion **222** in the through hole **110**. Though the receiving portion **150** has the ring shape while the held portion **222** has a flange shape in the connector **1** of FIG. 5, the present invention is not limited thereto. Provided that the held portion **222** is brought into contact with the receiving portion **150** and is held by the shell block **10**, it can have various shapes.

[0046] As understood from FIGS. 3 and 7, the fixing plate **30** shown is fixed to the shell block **10** by laser welding. As a result, as shown in FIG. 3, welding portions **32** are formed. However, the fixing method of the fixing plate **30** to the shell block **10** is not limited thereto. For example, latches may be provided to the fixing plate **30**, and the latches may be set on the shell block **10**. Alternatively, the fixing plate **30** may be press-fitted into the shell block **10**, or the fixing plate **30** may be bonded to the shell block **10**. Furthermore, though the fixing plate **30** shown is fixed to a front surface **12** of the shell block **10**, it may be disposed in and fixed to the large aperture portion **120** of the through hole **110**.

[0047] As shown in FIGS. 4 and 10, the fixing plate **30** has a main back surface **34**. As understood from FIGS. 7 and 10, when the fixing plate **30** is fixed to the shell block **10**, the main back surface **34** is brought into contact with the front surface **12** of the shell block **10** and is pressed against the waterproof gasket **40**. The main back surface **34** is pressed against the waterproof gasket **40**, so that the potting resin **50** which is filled into the small aperture portion **130** of the connector **1** from behind is prevented from leaking forward, and water is prevented from entering the inside of the connector **1**. In the present embodiment, the main back surface **34** of the fixing plate **30** is a flat surface. However, the main back surface **34** may be provided with a protruding portion protruding backward. In that case, the protruding portion may have a size and a shape which are suitable to be pressed against the waterproof gasket **40**. The protruding portion may have a ring shape to overlap the waterproof gasket **40** when seen in the front-rear direction, for example.

[0048] As shown in FIG. 8, the contact assembly **20** is provided with temporary holding portions **27**. As understood from FIGS. 5 and 8, in the connector **1** shown, the temporary holding portions **27** are protrusions protruding forward from the insulation member **220**. The fixing plate **30** shown is provided with temporary held portions **36**. In the connector **1** shown, the temporary held portions **36** are notches. The temporary holding portions **27** temporarily hold the temporary held portions **36**, so that a relative position of the fixing plate **30** is fixed with respect to the contact assembly **20**. As understood from FIGS. 8 and 9, in the connector **1** shown, the protrusions or the temporary holding portions **27** of the insulation member **220** are press-fitted into the notches or the temporary held portions **36**, so that the relative position of the fixing plate **30** is fixed with respect to the contact assembly **20**. Accordingly, the fixing plate **30** is prevented from being rattled when the fixing plate **30** is fixed to the shell block **10**.

[0049] As shown in FIG. 6, the potting resin **50** is filled into the small aperture portion **130**. The small aperture portion **130** has an edge which is positioned at a rear end of the small aperture portion **130** in the front-rear direction. The potting resin **50** has a lower end and an upper end in the up-down direction. The lower end of the potting resin **50** does not extend rearward beyond the edge of the small aperture portion **130**. Furthermore, the upper end of the potting resin **50** shown does not extend rearward beyond the edge of the small aperture portion **130**. Namely, the potting resin **50** shown remains within the small aperture portion **130**. However, the present invention is not limited thereto, and the upper end of the potting resin **50** may extend rearward beyond the edge of the small aperture portion **130**. The potting resin **50** stops water entering the connector **1** through interfaces between each of the contacts **210** and the insulation member **220** holding the contacts **210** and an interface between the through hole **110** of the shell block **10** and the insulation member **220**.

[0050] As understood from FIGS. 3 and 5, the back shell **60** is fixed to the shell block **10** and covers a rear end of the shell block **10**. The back shell **60** has a shield effect to prevent interference

of electromagnetic energy from high-frequency signals transmitted on the contacts **210** to peripheral parts. In addition, the back shell **60** prevents entering of dust and foreign substances and deformation of the connection portion **214** (see FIG. **6**) made from rearward of the connector **1**. [0051] As described with reference to FIG. **7**, in the connector **1** of the present invention, the second size **S2** or the size of the tallest part of the rear portion **25** is smaller than the first size **S1** or the size of the shortest part of the small aperture portion **30**. In other words, the size of the tallest part of the rear portion **25** of the contact assembly **20** is smaller than the size of the shortest part of the through hole **110** of the shell block **10**. Accordingly, as understood from FIGS. **7**, **9** and **10**, the rear portion **25** having the connection portions **214** can be inserted into the through hole **110** of the shell block **10**. That is, the contact assembly **20** can be assembled with the shell block **10** from a front of the shell block **10**. Furthermore, as described with reference to FIG. **6**, this structure can sufficiently show a water stop effect even if the lower end of the potting resin **50** does not extend rearward beyond the edge of the small aperture portion **130**. Accordingly, a size of the connector **1** can be reduced in the length direction or the front-rear direction.

[0052] Although the specific explanation about the present invention is made above with reference to concrete embodiments, the present invention is not limited thereto but susceptible of various modifications and alternative forms without departing from the spirit of the invention. For example, the contacts **210** with which the contact assembly **20** is provided may be freely decided in number and in arrangement. Moreover, the contact assembly **20** may have a mid-plate positioned at the middle thereof in the up-down direction.

[0053] While there has been described what is believed to be the preferred embodiment of the invention, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such embodiments that fall within the true scope of the invention.

Claims

1. A connector comprising a shell block and a contact assembly, wherein: the shell block has a through hole and a receiving portion; the through hole pierces the shell block in a front-rear direction and has a large aperture portion and a small aperture portion; the large aperture portion is located forward of the small aperture portion in the front-rear direction and communicates with the small aperture portion; the large aperture portion is larger than the smaller aperture portion in a plane perpendicular to the front-rear direction; the through hole has a first size as a size of a shortest part of the small aperture portion in an up-down direction perpendicular to the front-rear direction; the receiving portion is provided on a boundary between the large aperture portion and the small aperture portion in the through hole and faces forward in the front-rear direction; the contact assembly comprises at least one contact and an insulator member which fixes and holds the contact; the insulator member has a held portion projecting in directions perpendicular to the front-rear direction; the contact assembly is divided by the held portion into a front portion located forward of the held portion and a rear portion located rearward of the held portion; the held portion is brought into contact with the receiving portion in the through hole; the contact assembly has a second size as a size of a tallest part of the rear portion in the up-down direction; and the second size is smaller than the first size.
2. The connector as recited in claim 1, wherein: the connector further comprises a fixing plate fixed to the shell block; the held portion is sandwiched between the fixing plate and the receiving portion of the shell block in the front-rear direction; and the contact assembly is fixed to the shell block.
3. The connector as recited in claim 2, wherein: the connector further comprises a waterproof gasket; the waterproof gasket has a ring shape; and the waterproof gasket is attached to the shell block so that the waterproof gasket surrounds the shell block.
4. The connector as recited in claim 3, wherein: the shell block has a front end; the waterproof

gasket surrounds the front end of the shell block and a vicinity of the front end; the fixing plate has a main back surface; and when the fixing plate is fixed to the shell block, the main back surface is brought into contact with the front end of the shell block.

5. The connector as recited in claim 2, wherein: the contact assembly is provided with a temporary holding portion; the fixing plate is provided with a temporary held portion; and the temporary holding portion temporarily holds the temporary held portion, so that a relative position of the fixing plate is fixed with respect to the contact assembly.

6. The connector as recited in claim 1, wherein: the connector is mounted on a circuit board arranged under the rear portion in the front-rear direction when used; the contact has a connection portion which is connected to a conductor on the circuit board when the connector is mounted on the circuit board; the connector further comprises a potting resin filled into the small aperture portion; the small aperture portion has an edge which is positioned at a rear end of the small aperture portion in the front-rear direction; the potting resin has a lower end in the up-down direction; and the lower end of the potting resin does not extend rearward beyond the edge of the small aperture portion.

7. The connector as recited in claim 1, wherein the connector further comprises a back shell which is fixed to the shell block and covers a rear end of the shell block.

8. The connector as recited in claim 1, wherein: the connector is provided in a case of an electronic device when used; and the shell block is provided with a fixed portion fixed to the case of the electronic device.
